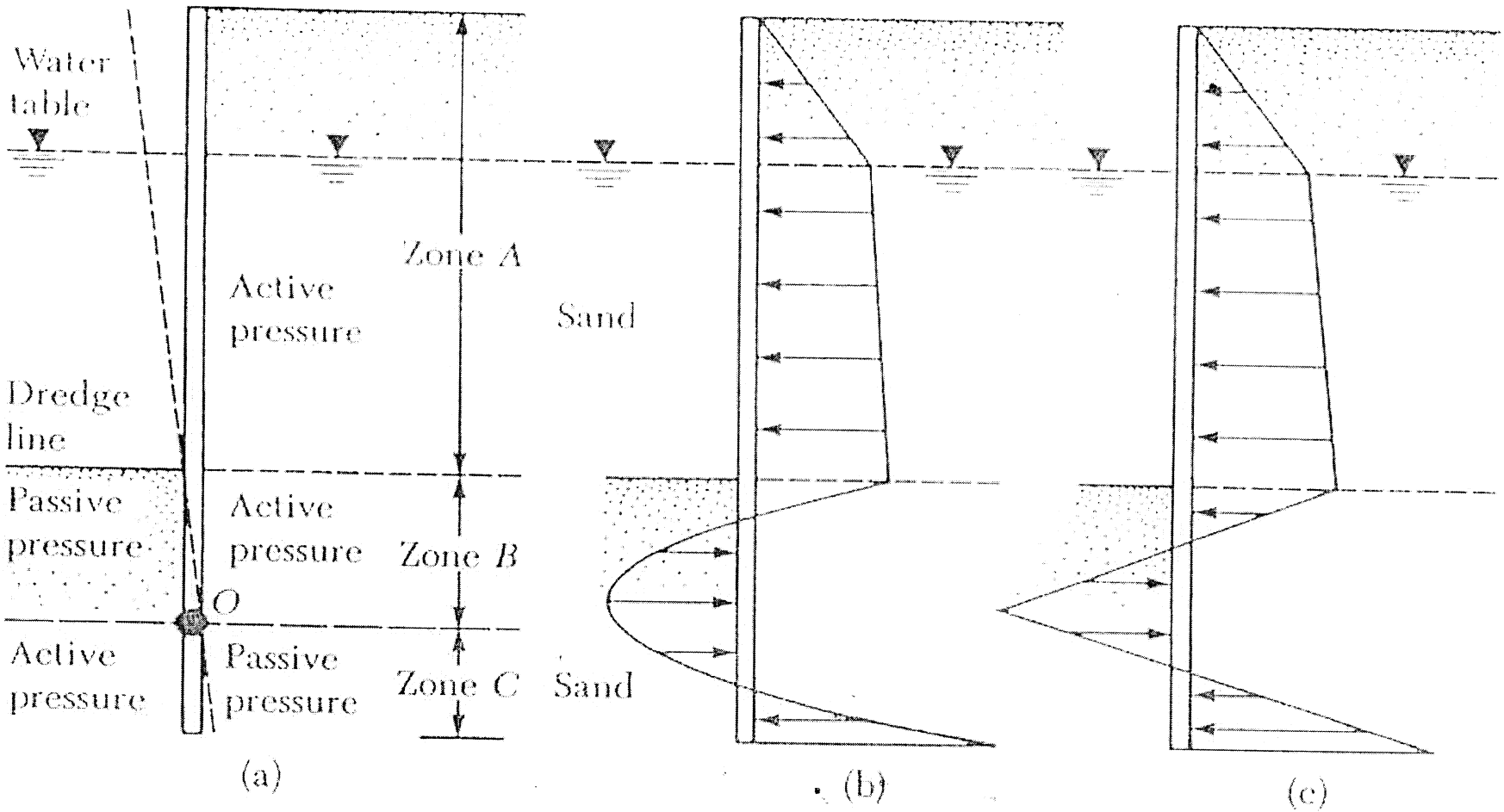
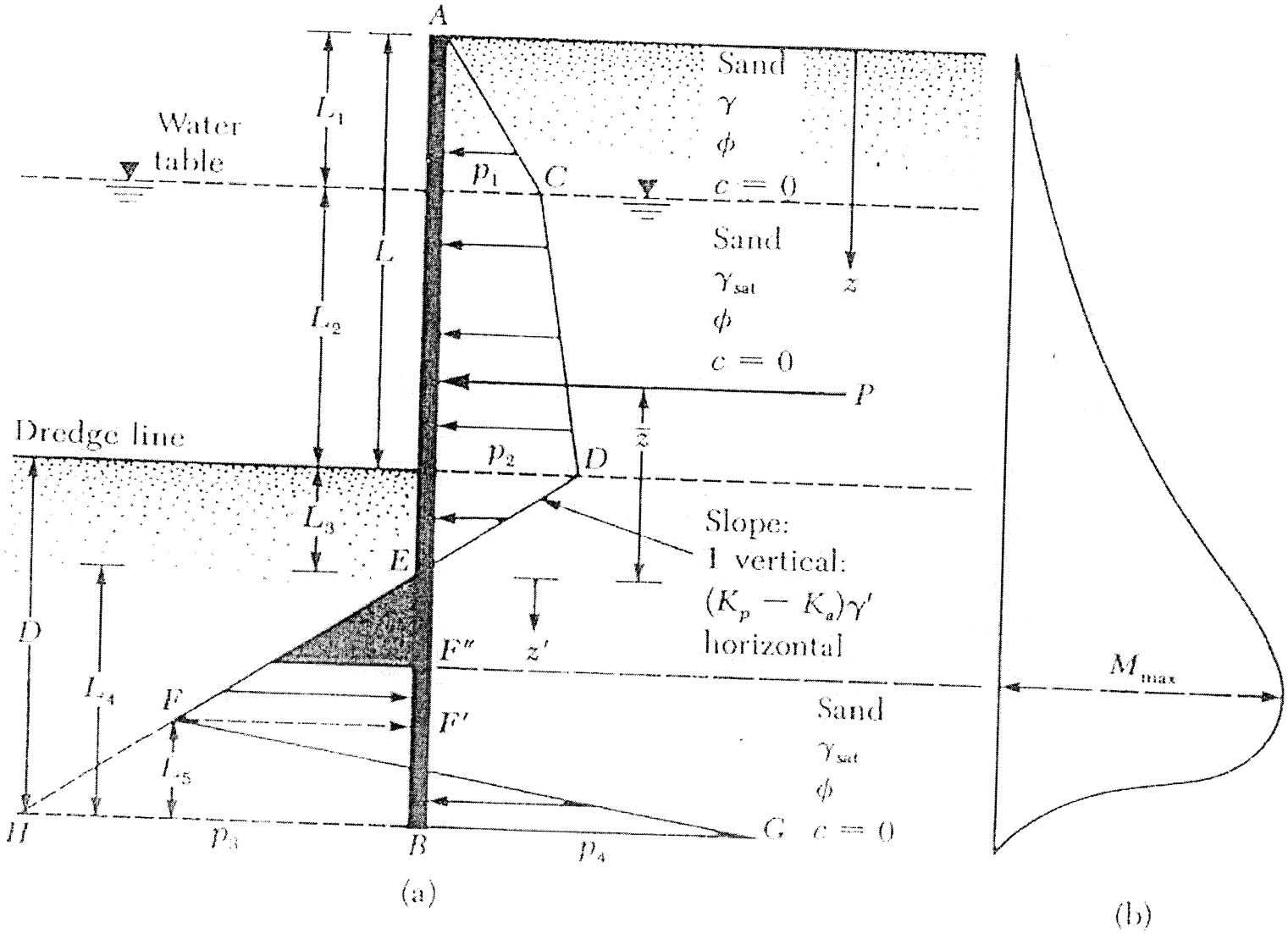
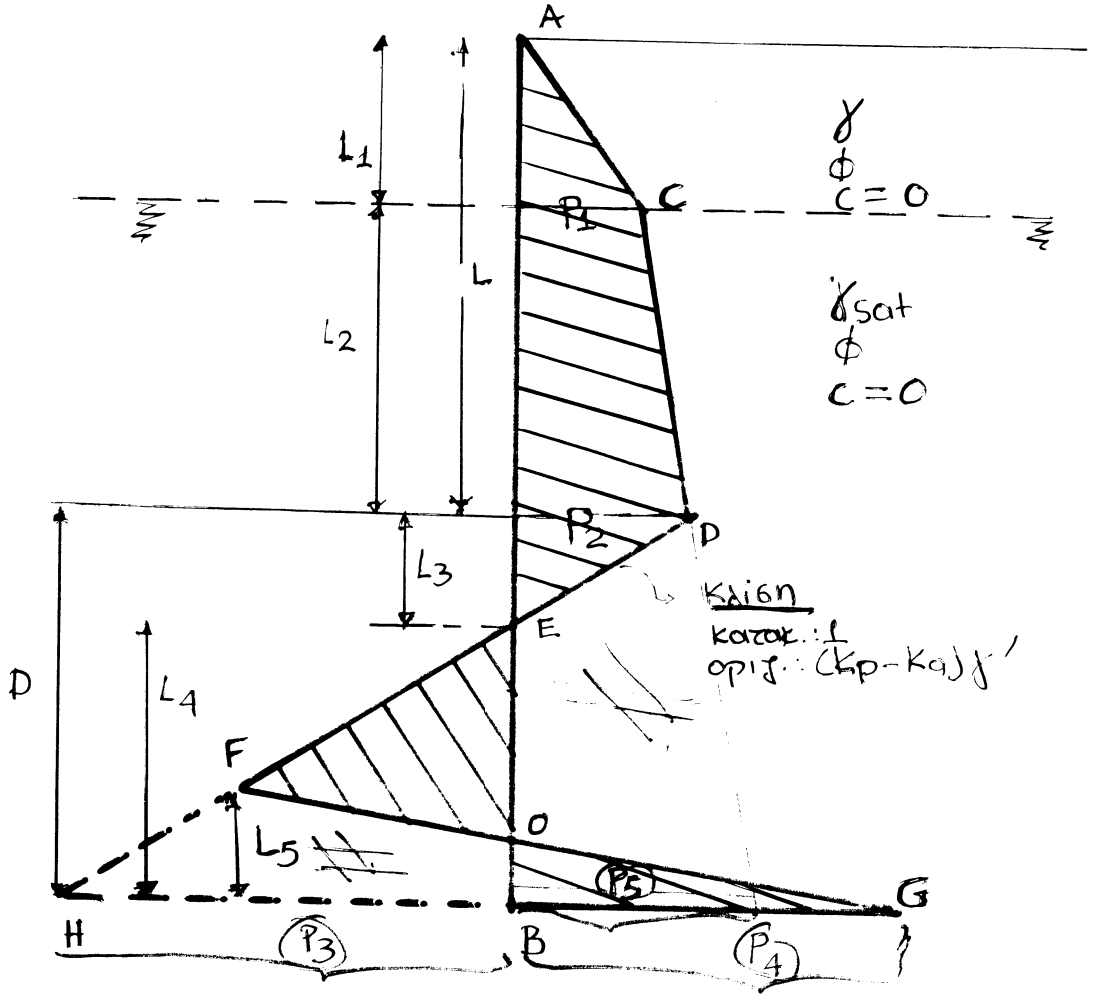


Das B.M.: Foundation Engineering



Das B.M.: Foundation Engineering


A
L_1
P_1
C
γ
ϕ
$c=0$
L
L_2
γ_{sat}
ϕ
$c=0$
P_2
D
L_3
E
Kdign
kazak.: 1
opis.: $(K_p-K_a)\gamma'$
D
L_4
F
O
L_5
P_5
G
H
P_3
B
P_4

$L_5 = z$

Άσκηση: Αντιστήριξη με παραλοβανίδες σε αμμώδες έδαφος

Δεδομένα: $L_1 = 2.0$ $L_2 = 2.4$ $\gamma = 17\ kN/m^3$ $\gamma_{sat} = 19\ kN/m^3$

$\phi = 32^\circ \rightarrow$ $K_a = 0.31$ $K_p = 3.25$

(βλ. Διάγραμμα συμβολισμών)

Να προσδιορισθεί το απαιτούμενο μήκος έμπηξης, το διάγραμμα ωθήσεων, η μέγιστη καμπτική ροπή και η απαιτούμενη ροπή αντίστασης της παραλοβανίδας.

Επίλυση

$\Sigma F_x = 0 \rightarrow$

$$0.5 L_1 P_1 + \frac{P_1+P_2}{2} L_2 + \frac{P_2+P_5}{2} D + \frac{P_3+P_4}{2} z - \frac{P_3+P_5}{2} D = 0 \rightarrow$$

$$z = \left[(P_3+P_5) D - L_1 P_1 - (P_1+P_2) L_2 - (P_2+P_5) D\right] / (P_3+P_4)$$

$P_1 = \gamma L_1 K_a = 17\ 2\ 0.31 = 10.54$

$P_2 = P_1 + \gamma' L_2 K_a = 10.54 + 9\ 2.4\ 0.31 = 17.2$

$P_3 = \gamma' D (K_p - K_a) - P_2 = 9 D (3.25 - 0.31) - 17.2 = 26.46 D - 17.2$

$P_5 = P_2 + \gamma' D K_a = 17.2 + \gamma' D\ 0.31 = 17.2 + \frac{2.79}{5.27} D$

$P_4 = (L_1 \gamma + L_2 \gamma' + D \gamma') K_p - D \gamma' K_a = 180.7 + 26.46 D$

Έστω $D = 4.70$ m

$P_1 = 10.54$ $P_2 = 17.2$ $P_3 = 107.2$ $P_4 = 305.1$ $P_5 = 30.3$

$$L_5 = [646.3 - 21.1 - 66.6 - 221.8] / 412.3 = 0.82$$

Θα πρέπει $\Sigma M_B = 0$

$$0.5 L_1 P_1 (L_2 + D + \frac{L_1}{3}) + P_1 L_2 (D + \frac{L_2}{2}) + \frac{P_2 - P_1}{2} L_2 (D + \frac{L_2}{3})$$

$$+ P_2 D D/2 + \frac{P_5 - P_2}{2} D D/3 + \frac{P_3 + P_4}{2} L_5 \frac{L_5}{3} - \frac{P_3 + P_5}{2} D D/5 =$$

$81.86 + 149.2 + 44 + 190 + 48.2 + 46.2 - 506.2 = 53.3$

Διαφορά εις είδην του 10%.

Αύξηση D κατά 10% → $D = 5.15$

$P_1 = 10.54$ $P_2 = 17.2$ $P_3 = 119.1$ $P_4 = 317$ $P_5 = 31.6$

$$L_5 = [776.1 - 21.1 - 66.6 - 251.32] / 436.1 = 1.0$$

$$\Sigma M_B = 86.6 + 160.6 + 47.6 + 228.1 + 63.7 + 72.7 - 666 =$$

$$= -6.7$$

Λύση αποδεκτή

Διαγραμμα Πιέσεων.

$P_1 = 10.54 \quad P_2 = 17.2 \quad P_4 = 317$

$P_F = (3.25 - 0.31) \, 9 \, (D - L_5) - 17.2 = 92.6$

Ηέγεση Καρτεσιανή Ροπή

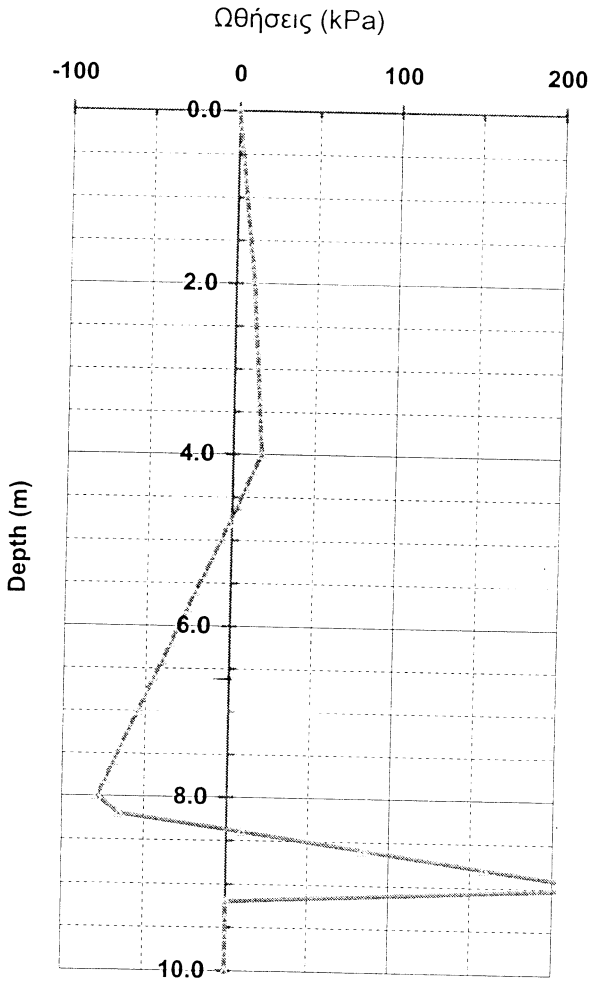
$F_x = 0 \rightarrow H_{max}$

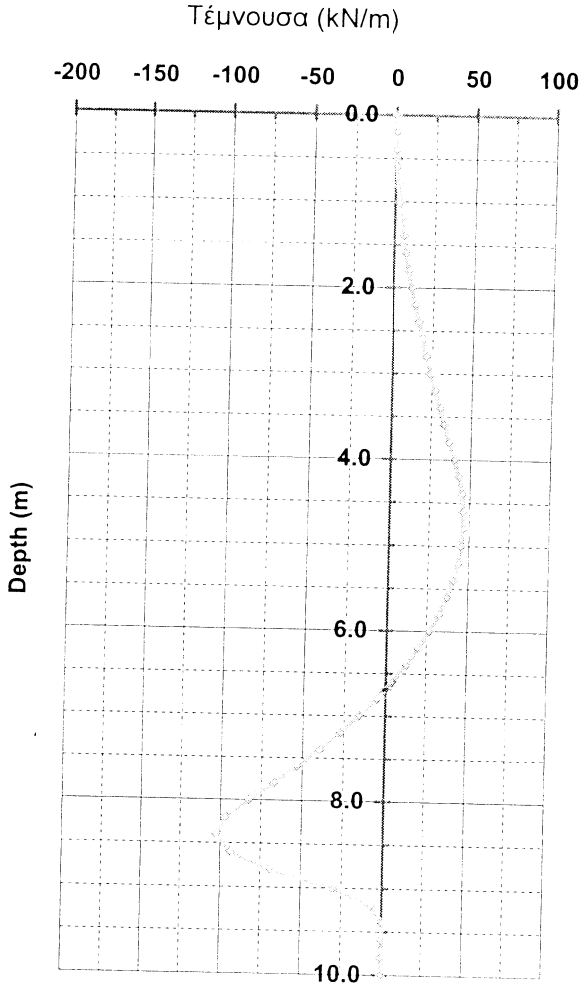
$$\left.\begin{aligned} \frac{L_1 P_1}{2} + \frac{P_1 + P_2}{2} L_2 + \frac{P_2 L_3}{2} &= (K_p - K_a) \gamma' z \frac{z}{2} \\ L_3 &= \frac{P_2}{(K_p - K_a)\gamma'} = 0.65 \end{aligned}\right\} z = 1.93$$

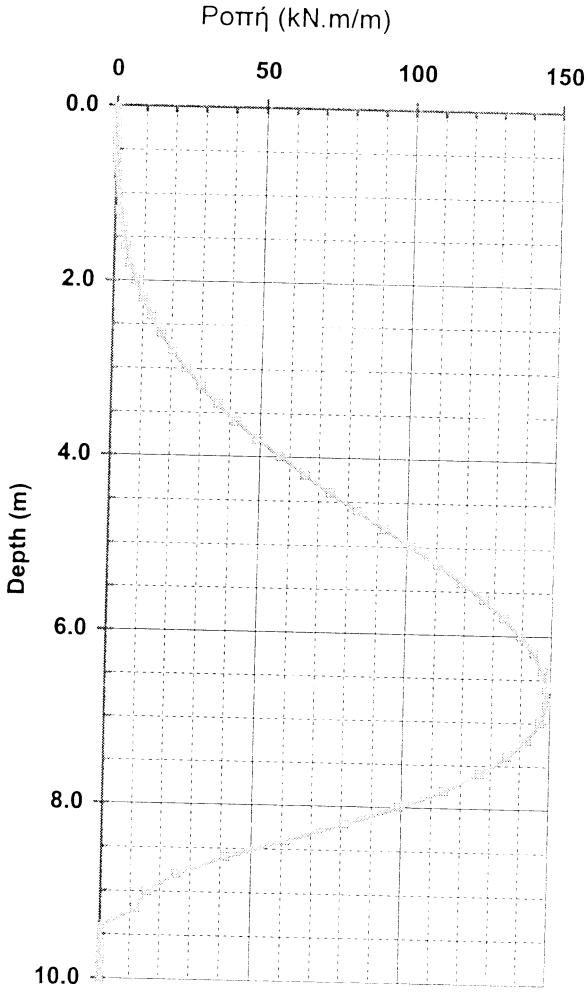
$$H_{max} = \frac{L_1 P_1}{2}\left(\frac{L_1}{3} + L_2 + L_3 + z\right) + P_1 L_2 \left(\frac{L_2}{2} + L_3 + z\right) + \frac{P_2 - P_1}{2} L_2 * \left(\frac{L_2}{3} + L_3 + z\right) + \frac{P_2}{2} L_3 \left(\frac{L_3}{3} z + z\right) - (K_p - K_a)\gamma' z \frac{z}{2} \frac{z}{3}$$

$= 59.5 + 95.6 + 27.0 + 13.2 - 31.7 = 163.6 \ KN.m$

$$\left.\begin{aligned} W &= \frac{H_{max}}{6} F \\ St\ 37 &\rightarrow 6 = 1.61 \ t/cm^2 \\ F &= 1.4 \end{aligned}\right\} W = \frac{1636}{1.61} \ t.cm \ 1.4 = 1422 \ cm^3$$







$\gamma = 17\ kN/m^3$

$\gamma_{sat} = 19$

$\phi = 30^\circ$

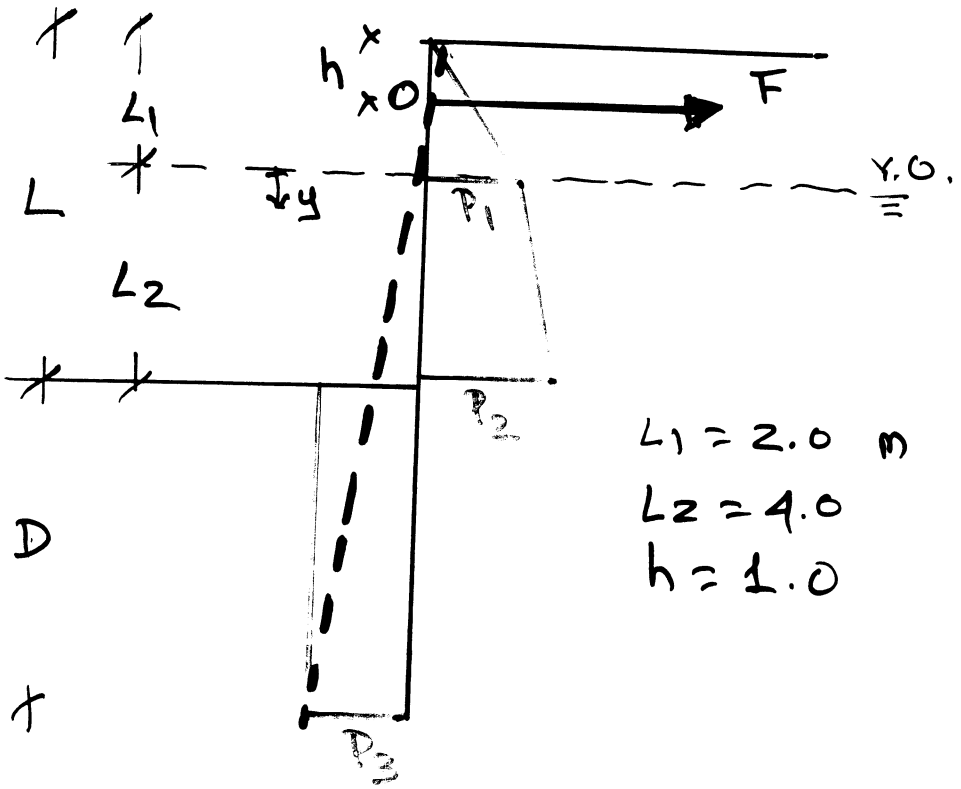
$L = 4.0$

$L_1 = 2.0$

$K_a = 0.33$

$K_p = 3.00$

ΔΙΑΦΡΑΓΜΑΤΙΚΟΣ ΤΟΙΧΟΣ ΑΠΟ ΠΑΣΣΑΛΟΣΑΝΙΔΕΣ
ΕΔΑΦΟΣ ΕΜΠΗΞΗΣ ΣΥΝΕΚΤΙΚΟ
ΑΓΚΥΡΩΣΗ



ΕΔΑΦΟΣ ΑΜΜΩΔΕΣ

$\gamma = 18$ $\gamma_{sat} = 20$

$\phi = 30^\circ$ — $K_a = 0.33$

$K_p = 3$

$L_1 = 2.0$ m

$L_2 = 4.0$

$h = 1.0$

ΕΔΑΦΟΣ ΣΥΝΕΚΤΙΚΟ

$C = 30$ kPa

$\gamma_{sat} = 21$ kN/m³

$K_a = K_p = 1$

$\Sigma F_x = 0$

$\Sigma H_O = 0$

$P_1 = L_1 \gamma K_a = 2.0 \cdot 18 \cdot 0.33 = 11.9$ kPa

$P_2 = P_1 + L_2 \gamma' K_a = 11.9 + 4 \cdot 10 \cdot 0.33 = 25.1$ kPa

$P_3 = \sigma_p^{\ell} - \sigma_h^{\ell} = \sigma_v' K_p + 2C - (\sigma_v' K_a - 2C)$

$= \gamma_c' D + 2C - L_1 \gamma - L_2 \gamma_s' + 2C$

$= 4C - L_1 \gamma - L_2 \gamma_s' = 120 - 2 \cdot 18 - 4 \cdot 10 = 44$ kPa

$$\Sigma H_O = 0 \rightarrow \frac{L_1 P_1}{2}\left(\frac{2}{3}L_1 - h\right) + P_2 L_2 \left(\frac{L_2}{2} + L_1 - h\right)$$

$$+ (P_2 - P_1)\frac{L_2}{2}\left(\frac{2}{3}L_2 + L_1 - h\right)$$

$$- P_3 D \left(\frac{D}{2} + L_2 + L_1 - h\right) = 0$$

$$\frac{2\ 11.9}{2}\left(\frac{2}{3}2-1\right)+11.9\ 4.0\left(\frac{4}{2}+2-1\right)$$

$$+(25.1-11.9)\frac{4}{2}\left(\frac{2}{3}4+2-1\right)$$

$$-4A\ D\left(\frac{D}{2}+4+2-1\right)=0$$

$$-22D^2-220D+243.6=0 \rightarrow D^2+10D-11.0=0$$

$$\rightarrow D=1.0\ m$$

$$\Sigma F_x=0 \rightarrow$$

$$\frac{P_1 L_1}{2}+\frac{P_1+P_2}{2}L_2-D\,P_3-F=0$$

$$F=\frac{11.9\ 2}{2}+\frac{25.1+11.9}{2}4-4A\ 1.0=41.9\ kN$$

$$\Sigma F_x(y)=0 \rightarrow$$

$$\frac{P_1 L_1}{2}+P_1 y+(P_2-P_1)\frac{y}{2}-41.9=0$$

$$\frac{11.9\ 2}{2}+\underset{11.9}{P_1}y+(\underset{13.2}{P_2-P_1})\frac{y}{2}-41.9=0 \rightarrow$$

$$y=1.62\ m$$

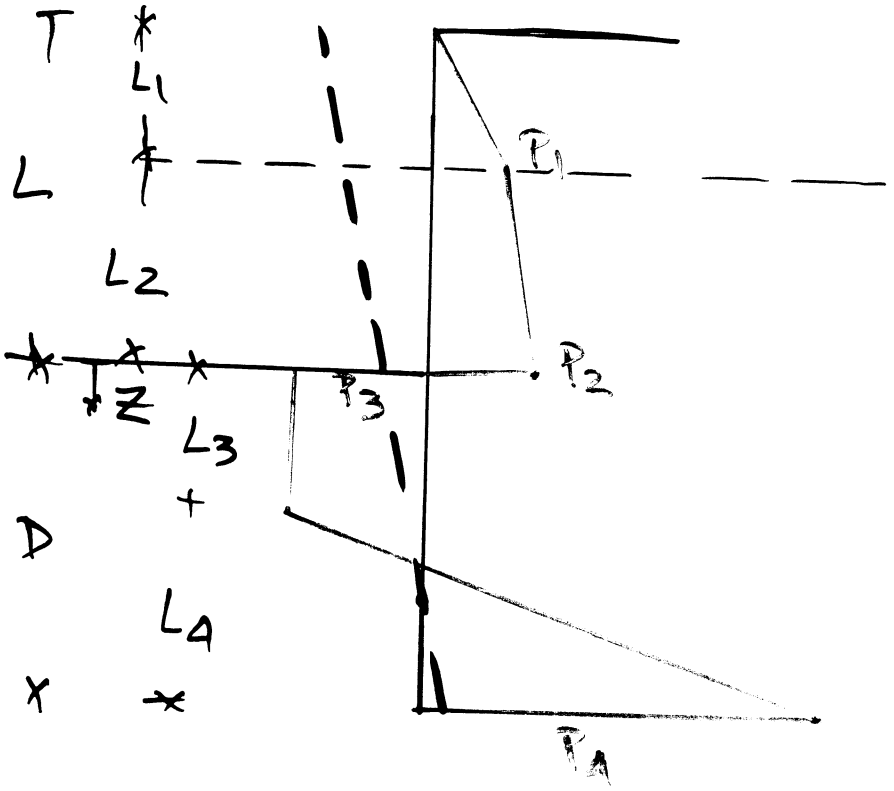
$$H_{max}=\frac{P_1 L_1}{2}\left(\frac{L_1}{3}+y\right)+\frac{P_1 y^2}{2}+\frac{P_2-P_1}{2}\frac{y^2}{3}-F(L_1+y-h)$$

$$=\frac{11.9\ 2}{2}\left(\frac{2}{3}+1.62\right)+\frac{11.9\ 1.62^2}{2}+\frac{13.2}{2}\frac{1.62}{3}-41.9\ 2.62$$

$$=11.9\ 2.29+15.4+5.77-110=-61.6\ kNm$$

$$W=\frac{M}{6}F=\frac{616}{1.61}1.4=535\ cm^3$$

ΔΙΑΦΡΑΓΜΑΤΙΚΟΣ ΤΟΙΧΟΣ ΧΩΡΙΣ ΠΑΣΣΑΛΟΣΑΝΙΔΕΣ ΜΕ ΑΚΤΥΡΩΣΗ



ΕΔΑΦΟΣ ΑΜΜΩΔΕΣ

$\gamma = 15$ $\gamma_{sat} = 19$ kN/m³

$\phi = 32 \rightarrow$ $k_a = 0.30$

ΕΔΑΦΟΣ ΣΥΝΕΚΤΙΚΟ

$\gamma_c^{sat} = 20$ kN/m³

$C = 50$ kPa

$L_1 = 2.0$ $L_2 = 3.5$

$P_1 = \gamma' L_1 k_a = 15\ 2\ 0.3 = 9.2$ kPa

$P_2 = P_1 + \gamma' L_2 k_a = 9.2 + 9\ 3.5\ 0.3 = 18.7$ kPa

$P_3 = \sigma_p^l - \sigma_a^r = 2C + \sigma_v'^l - (-2C + \sigma_v'^r)$

$= 4C - \gamma' L_1 - \gamma' L_2 = 4C - 15\ 2 - 9\ 3.5 = 138.5$ kPa

$P_A = \sigma_p^r - \sigma_a^l = 2C + \sigma_v^r - (\sigma_v^l - 2C)$

$= 4C + L_1\ 18 + L_2\ 9 = 261.5$ kPa

$\Sigma F_x = 0 \rightarrow$

$$\frac{P_1 L_1}{2} + P_1 L_2 + \frac{P_2 - P_1}{2} L_2 - P_3 D + \frac{P_3 + P_4}{2} L_4 = 0$$

$$\frac{9.2 \cdot 2}{2} + 9.2 \cdot 3.5 + \frac{9.5}{2} 3.5 - 138.5 D + 200 L_4 = 0$$

$$9.2 + 32.2 + 16.6 - 138.5 D + 200 L_4 = 0 \rightarrow$$

$$L_4 = 0.69 D - 0.29$$

Εστω $D = 2.5 \rightarrow L_4 = 1.44$

$\Sigma H_b = 0 \rightarrow$

$$\frac{P_1 L_1}{2} \left(\frac{L_1}{3} + L_2 + D\right) + P_1 L_2 \left(\frac{L_2}{2} + D\right) + \frac{P_2 - P_1}{2} L_2 \left(\frac{2}{3} L_2 + D\right) - P_3 \frac{D^2}{2} + \left(\frac{P_3 + P_4}{2}\right) L_4 \frac{L_4}{2} = 0$$

πρέπει

$$\frac{9.2 \cdot 2}{2} \left(\frac{2}{3} + 3.5 + 2.5\right) + 9.2 \cdot 3.5 \left(\frac{3.5}{2} + 2.5\right)$$

$$+ \frac{9.5}{2} 3.5 \left(\frac{2}{3} 3.5 + D\right) - 138.5 \frac{2.5^2}{2} + 200 \frac{1.44^2}{2} =$$

$$9.2 \quad 6.67 + 136.9 + 61 - 432.8 + 207 = 34.6$$

Έστω $D = 3 \rightarrow L_4 = 1.78$

$\rightarrow \Sigma M_b = -18.7$

$\rightarrow D \simeq 2.8 \rightarrow L_4 = 1.64$

Μέγιστη Ροπή

$\Sigma F_x (y) = 0$ (y από τη στάθμη του πυθμένα)

$$\frac{P_1 L_1}{2} + \frac{P_1 + P_2}{2} L_2 + P_3 y = 0$$

$$\rightarrow y = \frac{9.2 + 14 \times 35}{138.5} = 0.42$$

$$M_{max} = \frac{P_1 L_1}{2} \left(\frac{L_1}{3} + L_2 + y \right) + P_1 L_2^2 \left(\frac{L_2}{2} + y \right)$$

$$+ \frac{P_2 - P_1}{2} L_2 \left(\frac{L_2}{3} + y \right) - \frac{P_3 y^2}{2}$$

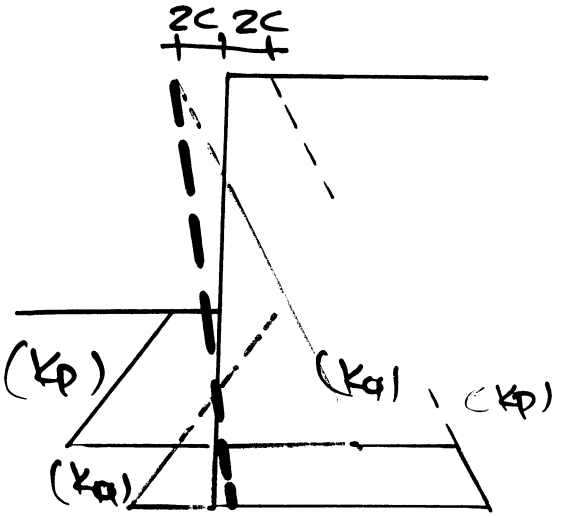
$$= 9.2\ 4.6 + 70 + 52.8 - 12.2 = 160\ kN.m$$

$$W = \frac{M_{max}}{6} \quad F = \frac{1600}{1.61} 1.4 = 1391\ cm^3$$

ΔΙΑΦΡΑΓΜΑΤΙΚΟΣ ΤΟΙΧΟΣ ΑΠΟ ΠΑΣΣΑΛΟΣΑΝΙΔΕΣ

ΕΔΑΦΟΣ ΣΥΝΕΚΤΙΚΟ - ΧΩΡΙΣ ΑΓΚΥΡΩΣΗ

ΕΙΚΟΝΑ ΠΑΡΑΜΟΡΦΩΣΗΣ - ΘΛΙΨΕΙΣ



$\sigma_h = \sigma_v K_a - 2c\sqrt{K_a}$

$\sigma_p = \sigma_v K_p + 2c\sqrt{K_p}$

$\phi = 0 \rightarrow$

$K_p = K_a = 1$

$\sigma_h = \sigma_v \overset{(1)}{K_a} - 2c$

$\sigma_p = \sigma_v \overset{(1)}{K_p} + 2c$

$\sigma_h = \sigma_v - 2c$

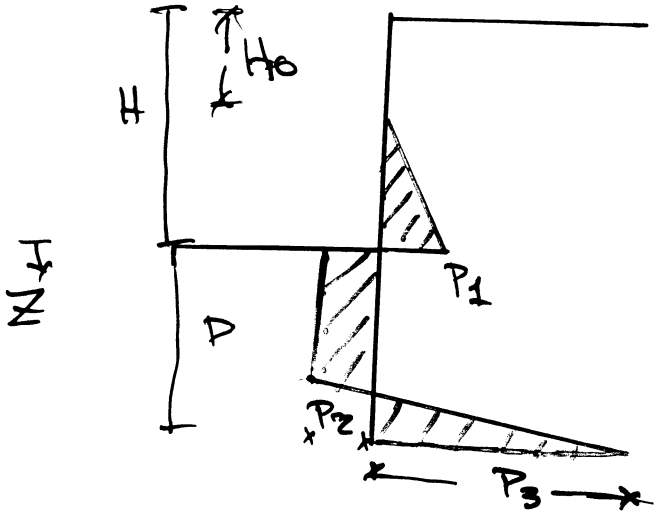
$\sigma_p = \sigma_v + 2c$

Διαγραμμα Θλίψεων

$\sigma_v = 0 \rightarrow \sigma_v = \gamma h$, $h = 0$

$\sigma_h = -2c$

$\sigma_p = +2c$



Κρίσιμο ύψος Hc
(αυτοσφήρωση εκσκαφή)

$\int_0^h \sigma_h = 0 \rightarrow \int \sigma_v - 2c = 0$

$\gamma \frac{h^2}{2} - 2ch = 0 \rightarrow$

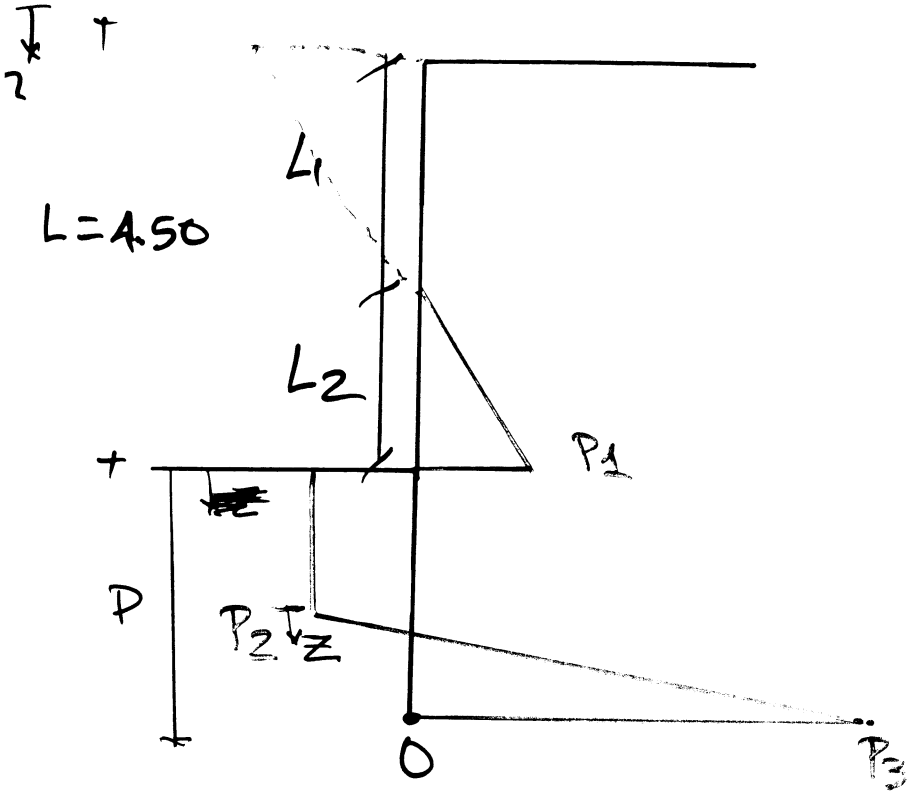
$h = \frac{4c}{\gamma} = H_c$

$P_1 = \sigma_h = \sigma_v - 2c = H\gamma - 2c$

$P_3 = \sigma_p^r - \sigma_h^l = \gamma(H+D) + 2c - \gamma D + 2c$

$= 4c + \gamma H$

$P_2 = \sigma_p^l - \sigma_h^r = \gamma z + 2c - \gamma H - \gamma z + 2c = 4c - \gamma H$



Δεδομένα

$\gamma = 19.2\ kN/m^3$

$c = 25\ kPa$

$\phi = 0 \rightarrow K_a = K_p = 1$

$\sigma_h = \sigma_r K_a - 2c = \sigma_r - 2c$

$\sigma_p = \sigma_r K_p + 2c = \sigma_r + 2c$

$L_2 + L_1 = L$ $\quad L_1$: Βάθος όπου $\sigma_h = 0$

$\sigma_h(h) = 0 \rightarrow \gamma h - 2c = 0 \rightarrow h = L_1 = \frac{2c}{\gamma} = \frac{50}{19.2} = 2.60$

$\rightarrow L_2 = 1.90m \rightarrow P_1 = (L_1 + L_2)\gamma - 2c$ $\quad \left.\begin{array}{l} P_1 = L_2\gamma = \\ 1.9\ 19.2 = 36.5 \\ kPa \end{array}\right\}$

$2c = \gamma L_1$

$P_2 = \sigma_p^l - \sigma_h^r = 4c - \gamma L = 13.6\ kPa$

$P_3 = \sigma_p^r - \sigma_h^l = 4c + \gamma L = 186.4\ kPa$

Προσδιορισμός Βάθους έμπηξης D

$\Sigma F_x = 0 \rightarrow \frac{P_1 L_2}{2} + P_2 D + (P_2 + P_3)\frac{x}{2} = 0$

Επαληθεύω

$\Sigma M_O = 0 \rightarrow \frac{P_1 L_2}{2}\left(D + \frac{L_2}{3}\right) - P_2 D\frac{D}{2} + (P_2 + P_3)\frac{x}{2}\frac{x}{3} = 0$

Επιλογή D → Z έλεγχος ροπών.

Εστω $D = 4.5m \rightarrow$

$$\frac{36.5\ 1.90}{2} - 13.6\ 4.5 + 200\ \frac{Z}{2} = 0 \rightarrow Z = 0.26$$

$$\Sigma H_o = \frac{36.5\ 1.90}{2}\left(4.5 + \frac{1.90}{3}\right) - 13.6\ \frac{4.5^2}{2} + 200\ \frac{Z^2}{6} = 0$$

$$= 48.9 \text{ kN.m} \neq 0$$

Εστω $D = 5.0 \rightarrow Z = 0.33$

$\Sigma H_o =$ [illegible] kN.m αποδεκτή (ελαφρώς)

Θέση μέγιστης καμπτικής ροπής

$\Sigma F_x = 0 \qquad F_x(y) = 0$ (όπου y το βάθος από τον πυθμένα της εκσκαφής)

$$\frac{P_1 L_z}{2} - P_2\ y = 0 \rightarrow y = \frac{36.5\ 1.90}{2\ \ 13.6} = 2.54\ m$$

$$H_{max} = 36.5\ \frac{1.90}{2}\left(\frac{1.90}{3} + 2.54\right) - \frac{13.6\ 2.54^2}{2}$$

$$= 66 \text{ kN.m}$$